Local-to-Global Bayesian Computation

Abstract

This proposal develops a Bayesian computation framework for large structured model spaces. The main idea is to run fast local posterior computations on small subproblems, merge the resulting local structures, and then correct the merged output toward the global posterior distribution. The project is motivated by Bayesian variable selection, where merging local structures is combinatorially easy but posterior uncertainty is not preserved. The proposed framework treats local posterior samples as particles or proposals, not as the final posterior. Correction is achieved through importance weighting, sequential Monte Carlo, tempering, or Metropolis–Hastings steps targeting the full posterior. The goal is a methodology strong enough for a statistics journal such as *JASA* or *JRSSB*: a clear statistical principle, rigorous validity results, and convincing demonstrations in both familiar and challenging structured model spaces.

1 Motivation

Many Bayesian inference problems involve posterior distributions over large structured model spaces. Examples include variable-selection models, graphical models, partitions, and symbolic expressions. In such problems, the posterior has the general form

$$\pi(M, \theta_M \mid y) \propto L(y \mid M, \theta_M)\pi(M, \theta_M),$$

where $M \in \mathcal{M}$ is a structured model object and θ_M denotes the continuous parameters associated with M. The model object M may be a selected variable set, a graph, or another combinatorial structure.

Direct Markov chain Monte Carlo over (M, θ_M) may be slow for at least three reasons. First, the space $\mathcal{M}$ may be enormous. Second, the posterior may be multimodal. Third, likelihood evaluation can be expensive. A natural strategy is therefore to divide the problem into smaller local subproblems, run fast Bayesian computations locally, and combine the local results.

The central statistical difficulty is that naive local-to-global merging usually does not recover the full Bayesian posterior. Even if the merge operation is simple, posterior uncertainty can be wrong. This point is especially clear in Bayesian variable selection: local analyses can identify variables from different blocks, and taking their union is easy, but the resulting uncertainty may ignore cross-block dependence.

This project studies the following principle:

local posterior computation $\longrightarrow$ merge local structures $\longrightarrow$ correct toward the global posterior.

The correction step is essential. Local posterior samples are useful computational building blocks, but they are not the target posterior distribution.

2 General Framework

Let $M \in \mathcal{M}$ denote a structured model object and let $\theta_M \in \Theta_M$ denote its continuous parameters. The full posterior is

$$\pi(M, \theta_M \mid y) = \frac{L(y \mid M, \theta_M)\pi(M, \theta_M)}{Z}, \tag{1}$$

where Z is the normalizing constant.

Suppose the full problem can be decomposed into K local subproblems. For each local subproblem k, we run a local Bayesian computation and obtain samples

$$(M_k^{(s)}, \theta_k^{(s)}) \sim \widetilde{\pi}_k(M_k, \theta_k \mid y_k), \qquad k = 1, \ldots, K.$$

The local posterior $\widetilde{\pi}_k$ is usually a working posterior, not a marginal of (1). The project therefore treats the local samples as proposals or particles. A merging mechanism maps local structures into global candidates:

$$\mathcal{A} : (M_1, \theta_1), \ldots, (M_K, \theta_K) \mapsto (M, \theta_M).$$

This induces a proposal or preliminary approximation $q(M, \theta_M)$ on the global space. The central task is to correct q toward the true posterior π.

Definition 1 (Local-to-global posterior correction)**.** *A local-to-global posterior correction method consists of:*

(i) local posterior computations $\widetilde{\pi}_1, \ldots, \widetilde{\pi}_K$;

(ii) a merge proposal q induced by local samples and a merge map $\mathcal{A}$;

(iii) a correction mechanism whose target is either the full posterior π in (1), or a clearly defined approximation to π with quantifiable error.

Several correction mechanisms are possible.

2.1 Importance correction

If $q(M, \theta_M)$ can be evaluated or estimated, then importance sampling uses weights

$$w(M, \theta_M) = \frac{L(y \mid M, \theta_M)\pi(M, \theta_M)}{q(M, \theta_M)}.$$

The corrected empirical distribution is

$$\widehat{\pi}_M = \sum_{s=1}^{M} \bar{w}_s \delta_{(M^{(s)}, \theta^{(s)})}, \qquad \bar{w}_s = \frac{w_s}{\sum_{\ell=1}^{M} w_\ell}.$$

The main weakness is weight degeneracy when q is far from π. This is expected in high-dimensional or highly discrete model spaces.

2.2 Metropolis–Hastings correction

Alternatively, use the local-to-global mechanism as a proposal in a Metropolis–Hastings algorithm. Given current state $x = (M, \theta_M)$ and proposal $x' = (M', \theta'_{M'}) \sim q(x, \cdot)$, accept with probability

$$\alpha(x, x') = \min\left\{1, \frac{\pi(x' \mid y)q(x', x)}{\pi(x \mid y)q(x, x')}\right\}.$$

This gives exact posterior invariance when the proposal has adequate support and the posterior density can be evaluated up to proportionality.

Proposition 1 (Posterior invariance)**.** *Assume the proposal kernel $q(x, \mathrm{d}x')$ has support on the relevant posterior support and the Metropolis–Hastings acceptance probability is computed using the full posterior density in (1). Then the resulting Markov chain is reversible with respect to π, and hence π is invariant.*

Proof. This is the standard detailed balance argument. For $x \neq x'$,

$$\pi(x)q(x, x')\alpha(x, x') = \min\{\pi(x)q(x, x'), \pi(x')q(x', x)\},$$

which is symmetric in x and x'. Therefore detailed balance holds, and π is invariant. □

2.3 Sequential Monte Carlo correction

A more stable strategy is to define a sequence of intermediate targets

$$\pi_0, \pi_1, \ldots, \pi_T, \qquad \pi_T = \pi,$$

where π_0 is induced by local posterior computations and π_T is the full posterior. Particles are propagated through the sequence by importance weighting, resampling, and MCMC rejuvenation.

For example, a tempered path may be

$$\pi_\lambda(M, \theta_M) \propto \{L(y \mid M, \theta_M)\}^\lambda \{\widetilde{L}(y \mid M, \theta_M)\}^{1-\lambda} \pi(M, \theta_M), \qquad \lambda \in [0, 1], \tag{2}$$

where $\widetilde{L}$ is a local or composite likelihood. At $\lambda = 0$, the target is local or approximate; at $\lambda = 1$, the target is the full posterior.

Definition 2 (MCMC rejuvenation)**.** *An MCMC rejuvenation step at intermediate target π_t is a Markov transition kernel K_t satisfying*

$$\pi_t K_t = \pi_t.$$

It is applied after resampling to restore particle diversity without changing the target distribution.

Proposition 2 (SMC consistency, informal)**.** *Suppose $\pi_0, \ldots, \pi_T$ are well-defined probability distributions, the incremental weights are correctly computed, and each rejuvenation kernel K_t leaves π_t invariant. Under standard regularity conditions for sequential Monte Carlo samplers, the empirical particle measure converges weakly to π_T as the number of particles tends to infinity.*

This result follows from standard SMC sampler theory (Del Moral et al., 2006). The novelty of the proposed project is not the generic SMC consistency theorem; the novelty is the construction of useful local-to-global paths and merge proposals for structured Bayesian model spaces.

3 Application I: Bayesian Variable Selection

3.1 Model

Consider Gaussian linear regression

$$y = X\beta + \epsilon, \qquad \epsilon \sim N(0, \sigma^2 I_n),$$

with p predictors. Let $S \subset \{1, \ldots, p\}$ denote the active set. A spike-and-slab posterior has the general form

$$\pi(S, \beta_S, \sigma^2 \mid y) \propto L(y \mid S, \beta_S, \sigma^2) \pi(\beta_S \mid S, \sigma^2) \pi(S) \pi(\sigma^2).$$

When p is large, full posterior exploration over 2^p models can be difficult. A split-and-merge strategy partitions predictors into blocks $B_1, \ldots, B_K$, runs local Bayesian variable selection in each block, and merges selected variables. The local posterior in block k is a working posterior

$$\widetilde{\pi}_k(S_k, \beta_{S_k}, \sigma^2 \mid y, X_{B_k}), \qquad S_k \subset B_k.$$

3.2 Why naive merging fails

The natural merge is

$$\widehat{S} = \bigcup_{k=1}^{K} \widehat{S}_k.$$

This merge is combinatorially easy. However, it does not preserve the full posterior distribution.

The main reason is cross-block dependence. Suppose two predictors X_j and X_ℓ are highly correlated but assigned to different blocks. A local analysis of the block containing j may assign high inclusion probability to j, while a local analysis of the block containing ℓ may assign high inclusion probability to ℓ. The merged result may then suggest that both variables should be included. The full posterior, however, may place mass on two competing models:

$$S = \{j\} \quad \text{or} \quad S = \{\ell\},$$

with little mass on $S = \{j, \ell\}$. Thus naive merging can destroy joint posterior uncertainty.

This example shows that variable selection is not trivial for the proposed framework. Even when the merge operation is easy, the posterior uncertainty problem remains hard.

3.3 Proposed algorithm for variable selection

The proposed variable-selection algorithm is:

1. Partition predictors into possibly overlapping blocks.
2. Run local Bayesian variable-selection chains in each block.
3. Use local posterior samples to build a global model proposal $q(S)$.
4. Extend S to (S, β_S, σ^2) by sampling or approximating the conditional posterior of continuous parameters.
5. Correct the global proposal using importance weighting, Metropolis–Hastings, or an SMC path ending at the full posterior.

A useful SMC bridge is

$$\pi_\lambda(S, \beta_S, \sigma^2) \propto \{L(y \mid S, \beta_S, \sigma^2)\}^\lambda \{\widetilde{L}(y \mid S, \beta_S, \sigma^2)\}^{1-\lambda} \pi(S, \beta_S, \sigma^2),$$

where $\widetilde{L}$ summarizes the local block analyses. The correction path moves from local evidence to the full likelihood.

3.4 Scientific role of variable selection

Variable selection is the controlled test bed for the project. It is a familiar topic, easy to simulate, and suitable for theory. Its role is to demonstrate the key statistical message:

$$\text{easy merge} \not\Rightarrow \text{valid posterior uncertainty}.$$

This application should be used to develop the core algorithm and to produce the first rigorous results.

4 Main Methodological Questions

The project should be organized around the following questions.

4.1 Question 1: What is the target?

For each algorithm, the target distribution must be explicit. The strongest version targets the true full posterior π. If exact targeting is too expensive, the approximation must be clearly defined and the error must be studied.

4.2 Question 2: What is the merge proposal?

The merge proposal is problem-specific. In variable selection, it may be a distribution over variable sets constructed from local inclusion probabilities. For other structured model spaces, it may be a problem-specific distribution over globally valid configurations.

4.3 Question 3: How is posterior uncertainty corrected?

The correction layer is the core statistical contribution. Possible correction methods include:

1. full-posterior Metropolis–Hastings correction;
2. sequential Monte Carlo with intermediate targets;
3. importance tempering from local to global posterior;
4. delayed acceptance using cheap local scores followed by full likelihood correction.

4.4 Question 4: When is correction stable?

Correction is only useful when the local proposal has enough overlap with the global posterior. The project should therefore study weight degeneracy, effective sample size, cross-block dependence, and sensitivity to block construction.

5 Potential Theoretical Results

A strong journal paper needs theory beyond an algorithm. The following results are natural targets.

5.1 Exactness under Metropolis–Hastings correction

As shown earlier, if local-to-global proposals are used inside a valid Metropolis–Hastings algorithm with full posterior ratios, then the full posterior is invariant. This result is simple but important because it separates proposal construction from posterior validity.

5.2 SMC consistency

For a well-defined sequence of intermediate targets ending at the full posterior, standard SMC theory gives convergence as the number of particles increases. The paper should state this formally for the proposed local-to-global construction.

5.3 Weight stability for variable selection

For variable selection, we can study when local proposals produce stable correction weights. A possible direction is to show that, under weak cross-block correlation, the discrepancy between a product of block posteriors and the full posterior is controlled by cross-block dependence terms.

A possible informal statement is:

Theorem 1 (Possible direction: weak cross-block dependence)**.** *Consider Gaussian linear regression with a sparse prior on S. Suppose the design matrix can be partitioned into blocks such that the cross-block correlations are uniformly small. Then, under suitable sparsity and signal-strength conditions, the divergence between the local block proposal and the full posterior is controlled by the magnitude of the cross-block covariance. Consequently, the variance of importance weights remains bounded when cross-block dependence is sufficiently weak.*

The exact assumptions and proof are part of the student's task. This theorem would formalize when local variable-selection proposals are useful.

5.4 Approximation error when exact correction is unavailable

If exact correction is computationally prohibitive, the paper should quantify approximation error. Possible metrics include

$$\mathrm{KL}(\pi \| \widehat{\pi}), \qquad \|\pi - \widehat{\pi}\|_{\mathrm{TV}},$$

or problem-specific summaries such as posterior inclusion probability error, joint inclusion probability error, and credible-interval coverage error.

6 Simulation Plan

6.1 Variable-selection simulations

Begin with Gaussian linear regression. Use settings such as

$$n \in \{100, 200, 500\}, \qquad p \in \{100, 500, 1000\}, \qquad s \in \{5, 10, 20\}.$$

Vary the following:

1. within-block correlation;
2. cross-block correlation;
3. signal strength;
4. block size;
5. overlap size;
6. prior sparsity.

Compare:

1. full posterior MCMC;
2. naive split-and-merge;
3. product of local posterior approximations;
4. one-shot importance correction;
5. tempered SMC correction;
6. block-informed Metropolis–Hastings correction.

Metrics:

1. marginal posterior inclusion probability error;
2. joint inclusion probability error;
3. posterior model-size distribution;
4. coefficient credible interval coverage;
5. prediction interval coverage;
6. effective sample size;
7. runtime.

The key demonstration should be a setting where naive split-and-merge screens variables well but gives wrong posterior uncertainty, especially for correlated predictors placed in different blocks.

7 Development Plan

7.1 Phase 1: Literature review

The student should first write a literature memo organized by five themes:

1. split-and-merge Bayesian variable selection;
2. composite likelihood and approximate posterior calibration;
3. divide-and-conquer Bayesian computation;
4. importance tempering and SMC samplers;
5. diagnostics for scalable Bayesian computation.

The memo should answer:

1. Which methods split data, variables, or model structures?
2. Which methods target the exact posterior?
3. Which methods target an approximation?
4. Which methods give valid posterior uncertainty?
5. Where is the gap for local-to-global structured model spaces?

7.2 Phase 2: Variable-selection prototype

The first implementation should be in Gaussian regression. The student should construct a simple example where cross-block correlation causes naive merging to distort uncertainty. Then the student should implement one correction method, preferably either Metropolis–Hastings correction or a simple tempered SMC correction.

7.3 Phase 3: First theorem

The first theorem should be modest and clean. The best options are:

1. exact posterior invariance under full-posterior Metropolis–Hastings correction;
2. SMC consistency for a specified local-to-global target sequence;
3. a weight-stability result under weak cross-block dependence in Gaussian regression.

7.4 Phase 5: Integrated paper

The paper should be written around the general principle and use variable selection as the main application. The experiments should demonstrate across several dependence regimes that easy merging is not enough for valid uncertainty.

8 Risks and Responses

8.1 Risk 1: The project is too broad

The framework could become too broad if it covers too many structured model spaces at once. The response is to keep the general framework simple and use variable selection as the controlled application:

variable selection: easy merge, hard uncertainty.

8.2 Risk 2: Novelty is challenged

A reviewer may say that the project combines known ideas: split-and-merge, SMC, and Metropolis–Hastings. The response is that the contribution is a general posterior-correction framework for local model-space exploration, with explicit attention to uncertainty distortion after merging.

To make the novelty strong, the paper should include at least one new algorithmic device, such as:

1. adaptive local-to-global tempering;
2. a proposal learned from local posterior samples;
3. correction diagnostics based on effective sample size or local-global discrepancy;
4. theory for stable correction weights.

8.3 Risk 3: Exact correction is expensive

The paper should separate exact and approximate variants. The exact variant uses full-posterior Metropolis–Hastings or SMC targets ending at the full posterior. The approximate variant uses local-to-global correction with quantified approximation error.

8.4 Risk 4: Local proposals miss important posterior regions

If local computations do not cover important global posterior regions, correction cannot recover them well. The method should therefore use overlapping blocks, multiple random partitions, adaptive blocking, tempering, rejuvenation, and diagnostics for degeneracy.

9 Why This Could Be a JASA or JRSSB Paper

A strong paper should make a broad statistical point:

> Local Bayesian computations are often used to scale posterior inference, but naive local-to-global merging can badly distort posterior uncertainty. We develop a correction framework that uses local computations for speed while targeting or approximating the global posterior distribution.

The paper should aim for three contributions:

1. **Conceptual contribution.** Define posterior correction after local model-space exploration as a general Bayesian inference problem.

2. **Methodological contribution.** Develop a divide–merge–correct Monte Carlo framework using SMC, tempering, and Metropolis–Hastings correction.

3. **Application contribution.** Demonstrate the framework in Bayesian variable selection across a range of dependence structures and problem sizes.

The ideal paper will have a clean algorithm, a rigorous validity result, one familiar statistical example, and one high-impact scientific application.

10 First Four-Week Assignment

The student's first four-week task should be:

1. Read the core references listed below.

2. Write a two-page literature map.

3. Implement a toy Gaussian variable-selection example with predictor blocks.

4. Show at least one example where naive merging distorts posterior uncertainty.

5. Implement a simple correction, either importance weighting or Metropolis–Hastings.

The first checkpoint is to show the central statistical phenomenon:

local merging can preserve signal but distort uncertainty.

11 Core References

Split-and-merge and feature partitioning

- Song and Liang (2015): split-and-merge Bayesian variable selection for ultrahigh-dimensional regression.
- Gutierrez and Guhaniyogi (2022): dynamic feature partitioning for high-dimensional Bayesian regression.

Composite likelihood and approximate posterior calibration

- Ribatet et al. (2012): Bayesian inference from composite likelihoods and the need for posterior calibration.
- Frazier et al. (2023): synthetic likelihood asymptotics and adjustments.

Importance tempering and SMC

- Zanella and Roberts (2019): scalable importance tempering for Bayesian variable selection.
- Lindsten et al. (2017): divide-and-conquer with sequential Monte Carlo.
- Del Moral et al. (2006): general theory of sequential Monte Carlo samplers.

Efficient Bayesian regression computation

- Hahn et al. (2019): efficient sampling for Gaussian linear regression with arbitrary priors.

12 Bottom Line

The project should aim to establish the following principle:

> Merging local Bayesian outputs is easy to do but hard to make statistically valid.

Variable selection gives a controlled and familiar setting where the merge is easy but posterior uncertainty can be wrong. It can support a broad methodology for local-to-global Bayesian computation over structured model spaces while keeping the paper focused.

13 Other thoughts

1. We should prove a theorem where our estimates of posterior expectation of $h(\theta)$ converge to the theoretical ground truth.
2. Song & Liang's approach might yield problematic uncertainty quantification.
3. Let's focus on variable selection and treat how to partition the parameter space as a random variable to be inferred, corresponding to stochastic scheduling.

References

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